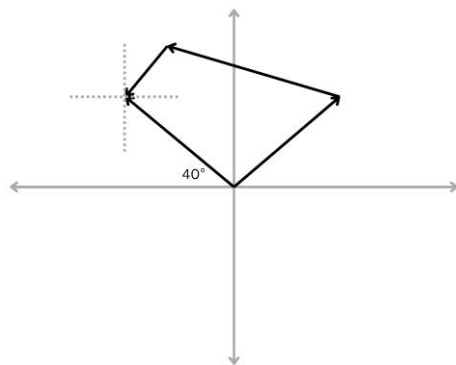


VECTOR ADDITION

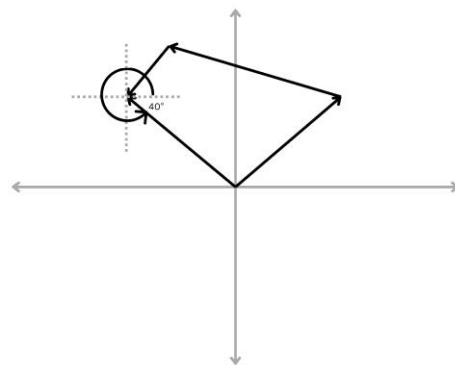
A vector quantity is represented by magnitude and direction. Force, displacement, and acceleration are examples of vectors. When performing operations involving vectors, the direction should also be considered. The sum of two or more vectors is called the resultant vector. We can add vectors using the polygon (graphical) or component (algebraic) method.

The resultant is the negative of the equilibrant, that is, they have equal magnitudes but opposite directions.



Resultant

Origin to Last Head



Equilibrant

Head-To-Tail

Figure 2.1. Sample resultant and equilibrant graphical representation.



Figure 2.2. Experimental setup for force table with hanging masses attached to the strings.

The force table has four pulleys for the four weights to be attached. Three will be used for the three forces then the other one for the equilibrant.

By trial and error, balance the three vectors by placing counterweights on the fourth unused string and locating the appropriate angle or direction. This is the Equilibrant.

Remark: The ring should be centered over the post when the system is in equilibrium. Pull the string slightly to one side and let it go. The ring should return to the center. If not, adjust the fourth vector's magnitude and direction until the ring returns to the center after pulling it to one side.

The **tail-to-tip method** is a way of adding vectors by drawing them one after another.

Each vector is like an arrow:

tail = starting point of the arrow

tip/head = pointed end of the arrow

In the tail-to-tip method:

1. Draw vector A first.
2. Place the **tail** of vector B at the **tip** of vector A.
3. Place the **tail** of vector C at the **tip** of vector B.
4. After all vectors are drawn, draw one arrow from the **original starting point** to the **last tip/head**. That arrow is the **resultant vector**.

So basically:

Resultant = from the first tail to the final tip

That is why it is called **tail-to-tip**: the tail of the next vector is connected to the tip of the previous vector.

SIGNIFICANT FORMULAS

Resultant = negative of Equilibrant

$$R = -E$$

Percentage Difference for Magnitude / Resultant

$$\%Difference_R = \frac{|R_2 - R_1|}{\frac{1}{2}(R_2 + R_1)} \times 100$$

Percentage Difference for Direction / Angle

$$\%Difference_\theta = \frac{|\theta_2 - \theta_1|}{\frac{1}{2}(\theta_2 + \theta_1)} \times 100$$